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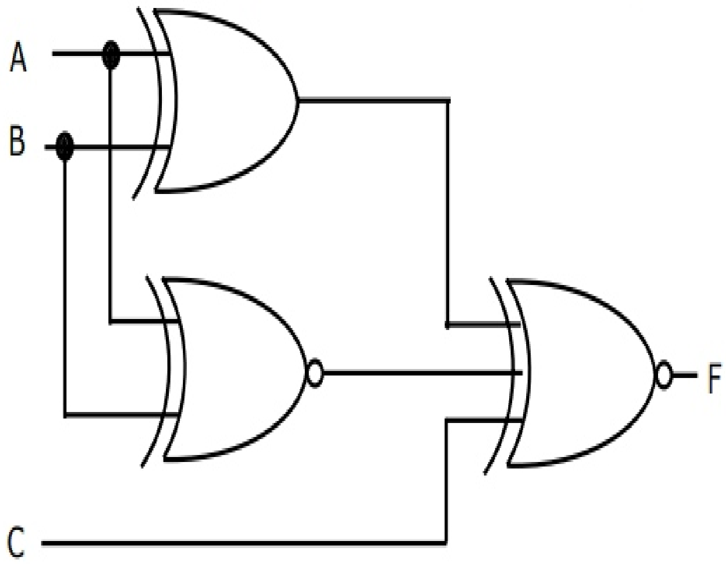
GATE 2010 EC, 12^{th} Question Analysis

Question 12

For the output F to be 1 in the logic circuit shown, the input combination should be



A A=1, B=1, C=0

B A=1, B=0, C=0

C A=0, B=1, C=0

D A=0, B=0, C=1

Question Analysis:

- For Two input XOR Gate: Let output be $X = A \oplus B$
- For Two input XNOR Gate: Let output be $Y = A \odot B$
- For Three input XNOR Gate: output $F = X \odot Y \odot C$
- The Final F expression is $F = (A \oplus B) \odot (A \odot B) \odot C$
- The Truth table for above expressions is as follows

The Truth Table

A	B	C	$X = A \oplus B$	$Y = A \odot B$	$F = X \odot Y \odot C$
0	0	0	0	1	0
0	0	1	0	1	1
0	1	0	1	0	0
0	1	1	1	0	1
1	0	0	1	0	0
1	0	1	1	0	1
1	1	0	0	1	0
1	1	1	0	1	1

Hardware Implementation

The above problem is implemented and tested in hardware using Arduino UNO board. Here by changing inputs A B C blinked the LED (output F is connected to LED) as per truth table and verified the expression.

Required Components & Pin Connections

S.No	Component
1	Arduino Uno Board
2	Breadboard
3	LEDs (1)
4	Resistors: 220Ω (1)
5	Jumper Wires
6	USB Cable

Component	Arduino Pin
Input A (5V or GND)	Digital 2
Input B (5V or GND)	Digital 3
Input C (5V or GND)	Digital 4
Output F (LED)	Digital 7
GND	GND
VCC	5V

Logic Description

- Let initialize inputs $A = 0, B = 0, C = 0$
- The output expressions of F from ciruit is as follows
- $F = (A \oplus B) \odot (A \odot B) \odot C$
- Now Change the input values as per Truth table and Note Down the Experimental Truth table as per LED status.

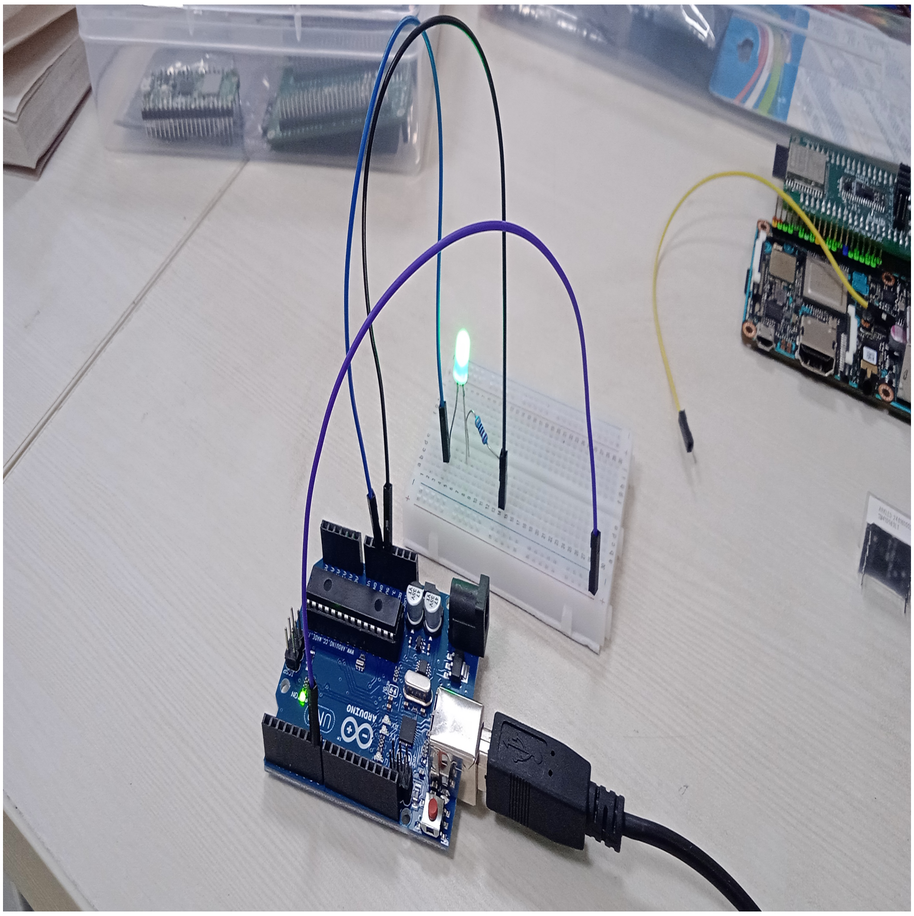
Code Uploading Steps

1. Create a Platform IO project
2. Write The code in main.cpp in src

3. Run the PIO project with command "pio run". It will compile the code and creates .hex file
4. Copy the .hex file to ArduinoDriod folder
5. connect the Arduino UNO to mobile with OTG cable
6. Upload the hex file using "upload precomplied" option
7. Observe the ouput and verify the expression

Experimental Truth Table

A	B	C	F (LED Output)
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1



Conclusion

- From Experimental Truth Table F will be 1 when $C = 1$.

- This matches option (D) from the original GATE question.
- The hardware experiment confirms the circuit's theoretical logic.